

1.

$$\phi(t) = e^{2(e^{it}-1)} \quad \dots \quad (1)$$

$$\phi(t) = e^{2(e^{it}-1)} \quad \dots \quad (2)$$

characteristic function.

$$\phi_X(t) = \mathbb{E}[e^{itX}]$$

a) $\phi(t) = \mathbb{E}[e^{i\omega t}]$ $\therefore \phi(0) = \mathbb{E}[e^0] = \mathbb{E}[1] = 1$

from (1) $\phi(0) = e^{2(e^0-1)} = 1$

(2) $\phi(0) = e^{2(e-1)} \neq 1$

$\therefore$ (2) can not be a characteristic function.

b) $\phi(t) = e^{2(e^{it}-1)} = \mathbb{E}[e^{itX}] = \int e^{itx} f_X(x) dx$

2. $X \sim \text{Nor}(Y, \sigma^2)$, $Y \sim \text{Exp}(\lambda)$

Find $M_{X+Y}(s)$

$$Z = \frac{X-Y}{\sigma} \sim N(0, 1)$$

$$M_Z(s) = e^{s^2/2}$$

$$X = \sigma Z + Y$$

$$M_X(s) = M_Z(\sigma s) M_Y(s)$$

$$M_Y(s) = \mathbb{E}[e^{sy}]$$

$$= \int_0^{\infty} e^{sy} \lambda e^{-\lambda y} dy = \frac{\lambda}{\lambda - s}$$

$$M_X(s) = \mathbb{E}[e^{sX}] = \frac{\lambda}{\lambda - 2s} = M_Y(2s)$$

$$\therefore X+Y = \sigma Z + 2Y$$

$$\Rightarrow M_{X+Y}(s) = M_Z(\sigma s) \cdot M_Y(2s) = e^{\sigma^2 s^2 / 2} \cdot \frac{\lambda}{\lambda - 2s}$$

$$3. M_X(s) = ae^s + be^{4(e^s-1)}, \quad \mathbb{E}[X] = 3$$

$$a) \frac{d}{ds} M_X(s) = ae^s + be^{4(e^s-1)} \cdot 4e^s$$

$$M_X(s)|_{s=0} = a+b=1$$

$$\frac{d}{ds} M_X(s)|_{s=0} = a + b \cdot 4 = 3$$

$$a + 4b = 3$$

$$\therefore 3b = 2 \quad b = \frac{2}{3}$$

$$a = 3 - 4b = 3 - \frac{8}{3} = \frac{1}{3}$$

$$b) P_X(1), \mathbb{E}[X^2], \mathbb{E}[X^3]$$

$$M_X(s) = \frac{1}{3}e^s + \frac{2}{3}e^{4(e^s-1)} \quad P(X=x) = \frac{1}{3} + \frac{2}{3} \frac{e^{-x} \lambda^x}{x!} \quad \lambda=4.$$

$$P(X=1) = \frac{1}{3} + \frac{2}{3} \cdot \frac{e^{-4} 4^1}{1!} = \frac{1}{3} + \frac{8}{3}e^{-4}$$

$$\mathbb{E}[X^2] = \frac{d^2}{ds^2} M_X(s)|_{s=0} = ae^s + 4e^s \cdot be^{4(e^s-1)} + 4e^s \cdot 4e^s \cdot be^{4(e^s-1)}|_{s=0}$$

$$= a + 4 \cdot b + 4 \cdot 4 \cdot b = a + 20b = \frac{1}{3} + \frac{40}{3} = \frac{41}{3}$$

$$\mathbb{E}[3^X] = \mathbb{E}[e^{X \ln 3}] = \mathbb{E}[e^{X \ln 3}]$$

$$= M_X(\ln 3)$$

$$\therefore \mathbb{E}[3^X] = \frac{1}{3}e^{\ln 3} + \frac{2}{3}e^{4(e^{\ln 3}-1)} = \frac{1}{3} \cdot 3 + \frac{2}{3} \cdot e^{4 \cdot 2} = 1 + \frac{2}{3}e^8$$

$$c) P(X+Y=2) = \sum_x P(X+Y=2|X=x)P(X=x)$$

$$= \sum_x P(Y=2-x|X=x)P(X=x) = \sum_x P(Y=2-x)P(X=x)$$

$$= P(Y=2)P(X=0) + P(Y=1)P(X=1) + P(Y=0)P(X=2)$$

$$= \left(\frac{1}{3} + \frac{2}{3} \cdot \frac{e^{-4} 4^2}{2!} \right) \left(\frac{1}{3} + \frac{2}{3} e^{-4} \right) + \left(\frac{1}{3} + \frac{2}{3} \frac{e^{-4} 4}{1!} \right) \left(\frac{1}{3} + \frac{2}{3} \frac{e^{-4} 4}{1!} \right) + \left(\frac{1}{3} + \frac{2}{3} e^{-4} \right) \left(\frac{1}{3} + \frac{2}{3} \cdot \frac{e^{-4} 4^2}{2!} \right)$$

$$4. M_X(s) = \frac{6-3s}{2(1-s)(3-s)}, \quad s < 1$$

$$= \frac{A}{1-s} + \frac{B}{3-s} = \frac{A(3-s) + B(1-s)}{(1-s)(3-s)}$$

$$3A+B=3, \quad -A-B=-\frac{3}{2}$$

$$\therefore 2A = \frac{3}{2}, \quad A = \frac{3}{4}, \quad B = 3 - \frac{3}{4} = \frac{9}{4}$$

$$\therefore M_X(s) = \frac{3/4}{1-s} + \frac{3/4}{3-s} = \frac{3}{4} \cdot \frac{1}{1-s} + \frac{1}{4} \cdot \frac{3}{3-s}$$

$$P(Z=X) = \frac{3}{4}, \quad P(Z=Y) = \frac{1}{4}$$

$$M_X(s) = \frac{1}{1-s} \therefore X \sim \text{Exp}(1) \quad M_Y(s) = \frac{3}{3-s} \therefore Y \sim \text{Exp}(3)$$

$$f_Z(z) = \frac{3}{4} e^{-z} + \frac{1}{4} e^{-3z} \quad F_Z(z) = \frac{3}{4}(1-e^{-z}) + \frac{1}{4}(1-3e^{-3z})$$

$$= 1 - \frac{3}{4}e^{-z} - \frac{3}{4}e^{-3z}, \quad z \geq 0$$

$$5. X_1, X_2 \sim \text{Exp}(1), \quad Y = X_1 - X_2$$

$$f_Y(z) = \frac{1}{2} e^{-|z|}$$

$$\varphi_Y(t) = \mathbb{E}[e^{itY}] = \mathbb{E}[e^{it(X_1-X_2)}] = \mathbb{E}[e^{itX_1}] \mathbb{E}[e^{-itX_2}] = \varphi_X(t) \varphi_X(-t)$$

$$= \frac{1}{1-it} \cdot \frac{1}{1+it} = \frac{1}{1-(it)^2} = \frac{1}{1+t^2}$$

$$6. X \sim \text{Bin}(200, \frac{1}{10})$$

Find upper bound of $P(X \geq 120)$

Markov inequality $P(X \geq t) \leq \frac{\mathbb{E}[X]}{t}$

$$\therefore P(X \geq 120) \leq \frac{20}{120} = \frac{1}{6}$$

Chebyshev inequality

$$P[|X-\mu| < k\sigma] \geq 1 - \frac{1}{k^2}$$

7. Gubner Chapter 3.3

$$G_X(z) = \left(\frac{2+z}{3}\right)^5 = E[z^X]$$

$$E[X] = \frac{d}{dz} G_X(z) \Big|_{z=1} = 5 \left(\frac{2+z}{3}\right)^4 \cdot \frac{1}{3} \Big|_{z=1} = \frac{5}{3}$$

$$E[X(X-1)] = \frac{d^2}{dz^2} G_X(z) \Big|_{z=1} = 20 \cdot \frac{1}{3} \cdot \frac{1}{3} \left(\frac{2+z}{3}\right)^3 \Big|_{z=1} = \frac{20}{9}$$

$$\begin{aligned} V[X] &= E[X^2] - E[X]^2 = E[X^2] - E[X] - E[X]^2 + E[X] \\ &= \frac{20}{9} - \frac{25}{9} + \frac{5}{3} = \frac{10}{9} \end{aligned}$$

8. Gubner Chapter 3.6

$$G_X(z) = (a_0 + a_1 z + a_2 z^2 + \dots + a_n z^n)^m / D$$

$$G_X(1) = E[1] = 1$$

$$= (a_0 + a_1 + a_2 + \dots + a_n)^m / D$$

$$\therefore D = (a_0 + a_1 + \dots + a_n)^m$$

9. Gubner Chapter 4.40

$$X \sim \text{Unif}(0,1) \quad Y = \ln(1/X) \sim \text{Exp}(1)$$

$$M_X(s) = E[e^{sx}] = \int_0^1 e^{sx} f(x) dx = \int_0^1 e^{sx} dx = s e^{sx} \Big|_0^1 = s(e^s - 1)$$

$$\begin{aligned} M_Y(s) &= E[e^{sY}] = E[e^{s \ln(1/X)}] = E[(1/X)^s] = E[1/X^s] \\ &= \int_0^1 \frac{1}{x^s} dx = \frac{1}{1-s} x^{-s+1} \Big|_0^1 = \frac{1}{1-s} \end{aligned}$$

$$\therefore Y \sim \text{Exp}(1)$$

10. Gubner, Chapter 4.66

$$f(x) = \frac{2}{x^3}, \quad x \geq 1 \quad \text{otherwise } f(x) = 0$$

$$\text{for } a \geq 1 \quad P(X \geq a)$$

Markov's inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

$$\mathbb{E}[X] = \int_1^{\infty} x \cdot \frac{2}{x^3} dx - \int_1^{\infty} \frac{2}{x^2} dx = -2 \cdot x^{-1} \Big|_1^{\infty} = 2$$

$$\therefore P(X \geq a) \leq \frac{2}{a}$$